

Web Data Integration
Project Outline

Pharmaceutical Drug Data

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1 Brief description of use case

2 How the datasets fulfill the requirements

2.1 Schema and basic profile of each dataset

Table 1 summarises the three input datasets. Table 2 lists the full attribute sets of Drugs@FDA and EMA Medicines.

Table 1: Overview of the input datasets

Dataset	Entities	Attributes	Data Format
Drugs@FDA	51,864 products	37	TXT/TSV (12 tables)
EMA Medicines	2,746 medicines	39	XLSX
openFDA NDC	138,209 products	10 (selected)	JSON → CSV

Drugs@FDA

Drugs@FDA contains **51,864 product records** across **29,377 applications**. The product table contains only **8,388 distinct drug names** and **3,355 distinct active-ingredient strings**, showing substantial reuse across products. Consequently, **Drug-Name + ActiveIngredient** provides a more appropriate starting point for entity linkage than DrugName alone.

The dataset is strongly imbalanced. Among applications, **78.3% are ANDAs**, **20.1% NDAs**, and **1.7% BLAs**. Among 194,004 submission records, **85.6% are supplemental**, while **99.4%** have submission status AP.

Three attributes exceed 30% missingness: **ApplPublicNotes (99.99%)**, **SubmissionsPublicNotes (99.83%)**, and **ReviewPriority (41.73%)**. The first two are optional regulatory note fields and are therefore populated only in exceptional cases. Missing **ReviewPriority** values cannot be explained by submission type alone: missingness remains high for both original submissions (**49.5%**) and supplements (**40.4%**). Overall, **ReviewPriority** consists of **49.3% Standard**, **5.8% Priority**, **3.2% other codes**, and **41.7% missing values**.

EMA Medicines

The EMA dataset contains **2,746 medicine records** and **39 attributes**, with no duplicate rows. It is predominantly composed of human medicines (**85.6%**). Medicine status is also imbalanced: **68.5% are currently authorised**, **15.4% withdrawn**, and **10.4% represent withdrawn applications**. Fifty medicine names occur more than once, again showing that names alone do not uniquely identify entities.

Twelve EMA attributes exceed 30% missingness. Most of this is **structural rather than erroneous**. Veterinary attributes are inapplicable to human medicines, while

refusal, withdrawal, suspension, and rolling-review dates exist only if the respective regulatory event occurred. Two more broadly applicable attributes require closer inspection: Opinion adopted date is **46.2% missing**, and Revision number is **30.2% missing**.

openFDA NDC Directory

The openFDA National Drug Code (NDC) Directory contains product listing information submitted to the FDA by drug labelers. The dataset was downloaded in JSON format and converted to CSV. It contains 138,209 product records and 7,682 distinct active ingredient entries. The listing includes both FDA-approved and unapproved products. Inclusion in the directory does not imply FDA approval.

Attributes: product NDC, brand and generic names, active ingredients, manufacturer, dosage form, route, marketing start date, marketing category, and application number. Marketing start date is not necessarily the approval date.

Across the 10 selected columns, 113,976 of 1,382,090 values are missing, which is approximately 8.25%, assuming the remaining columns have no missing values. The missing values by column:

- **brand_name**: 21,716 (15.71%)
- **generic_name**: 3 (less than 0.01%)
- **active_ingredients**: 2,521 (1.82%)
- **manufacturer**: 28,052 (20.30%)
- **route**: 23,596 (17.07%)
- **application_number**: 38,088 (27.56%)

Attribute overview

Attributes marked with ****** have more than 30% missing values.

Table 2: Attributes of Drugs@FDA and EMA Medicines

Dataset	Source (*)	Format	# Entities	# Attributes	List of attributes (**)
Drugs@FDA	Official U.S. FDA Drugs@FDA Data Files. Downloaded and extracted into 12 relational text tables.	TXT/TSV 12 relational tables	51,864 products 29,377 applications	56 table-columns 37 unique names	ActionTypes_Lookup: ActionTypes_LookupID, ActionTypes_LookupDescription, SupplCategoryLevel1Code, SupplCategoryLevel2Code.

Dataset	Source (*)	Format	# Entities	# Attributes	List of attributes (**)
					ApplicationDocs: ApplicationDocsID, ApplicationDocsTypeID, ApplNo, SubmissionType, SubmissionNo, ApplicationDocsTitle, ApplicationDocsURL, ApplicationDocsDate. Applications: ApplNo, ApplType, ApplPublicNotes **, SponsorName. ApplicationsDocsType_ Lookup: ApplicationDocsType.Lookup_ ID, ApplicationDocsType_ Lookup_Description. Join_Submission_ ActionTypes_Lookup: SubmissionType, j_ submissionActionTypeID, ApplNo, SubmissionNo, ActionTypes.LookupID. MarketingStatus: MarketingStatusID, ApplNo, ProductNo. MarketingStatus_Lookup: MarketingStatusID, MarketingStatusDescription. Products: ApplNo, ProductNo, Form, Strength, ReferenceDrug, DrugName, ActiveIngredient, ReferenceStandard. SubmissionClass_Lookup: SubmissionClassCodeID, SubmissionClassCode, Submis- sionClassCodeDescription. SubmissionPropertyType: ApplNo, SubmissionType, SubmissionNo, SubmissionPropertyTypeCode, SubmissionPropertyTypeID.

Dataset	Source (*)	Format	# Entities	# Attributes	List of attributes (**)
					Submissions: ApplNo, SubmissionClassCodeID, SubmissionType, SubmissionNo, SubmissionStatus, SubmissionStatusDate, SubmissionsPublicNotes **, ReviewPriority *.
					TE: ApplNo, ProductNo, MarketingStatusID, TECode.

Dataset	Source (*)	Format	# Entities	# At-tributes	List of attributes (**)
EMA Medicines	Official European Medicines Agency “Download medicine data” dataset. Imported from the official XLSX file.	XLSX	2,746 medicines	39	Category; Name of medicine; EMA product number; Medicine status; Opinion status ^{**} ; Latest procedure affecting product information; International non-proprietary name (INN) / common name; Active substance; Therapeutic area (MeSH); Species (veterinary) ^{**} ; Patient safety; ATC code (human); ATCvet code (veterinary) ^{**} ; Pharmacotherapeutic group (human); Pharmacotherapeutic group (veterinary) ^{**} ; Therapeutic indication; Accelerated assessment; Additional monitoring; Advanced therapy; Biosimilar; Conditional approval; Exceptional circumstances; Generic; Orphan medicine; PRIME: priority medicine; Marketing authorisation developer / applicant / holder; European Commission decision date; Start of rolling review date ^{**} ; Start of evaluation date ^{**} ; Opinion adopted date ^{**} ; Withdrawal of application date ^{**} ; Marketing authorisation date; Refusal of marketing authorisation date ^{**} ; Withdrawal / expiry / revocation / lapse of marketing authorisation date ^{**} ; Suspension of marketing authorisation date ^{**} ; Revision number ^{**} ; First published date; Last updated date; Medicine URL.

(*) **Source.** Drugs@FDA was obtained from the official FDA Drugs@FDA Data Files download and extracted into its 12 relational tables. EMA medicine data was obtained from the official EMA downloadable medicine-data spreadsheet.

(**) **Missingness.** Attributes marked with ^{**} contain more than 30% missing values. High missingness does not necessarily indicate poor data quality: some attributes are optional,

category-specific, or populated only when a particular regulatory event occurs.

2.2 Integrated schema and overlap with input schemata

Table 3: Attribute intersection with the integrated schema

Integrated Attribute	Drugs@FDA	EMA Medicines	openFDA NDC
Active Substance Name	X	X	X
[Attribute 2]	[X]	[X]	[X]
[Attribute 3]	[X]	[X]	[X]
[Attribute 4]	[X]	[X]	[X]
[Attribute 5 (Must share!)]	[X]	[X]	[X]
[List Attribute (e.g. Brands)]	[X]	[X]	[X]

2.3 Why enough entities are likely contained in multiple datasets

Exact normalized name + active substance matching currently identifies **428 shared FDA-EMA entities**, while normalized-name overlap reaches **752**. The required $\geq 1,000$ **shared entities** has therefore not yet been demonstrated. The third dataset should consequently provide substantial overlap with FDA or EMA and preferably contain active-substance information together with additional shared regulatory attributes.

Sources

- U.S. Food and Drug Administration, *Drugs@FDA Data Files*.
<https://www.fda.gov/drugs/drug-approvals-and-databases/drugsfda-data-files>
- European Medicines Agency, *Download medicine data*.
<https://www.ema.europa.eu/en/medicines/download-medicine-data>
- U.S. Food and Drug Administration, *openFDA NDC Directory*.
<https://open.fda.gov/data/ndc/>